Tumor hypoxia and survival in lung adenocarcinoma

Salvatore Megna

1 Introduction

Lung adenocarcinoma is one of the main forms of non-small-cell lung cancer and is characterized by substantial clinical and biological heterogeneity among patients. This heterogeneity can translate into important differences in disease progression, treatment response and survival. To investigate these aspects, a dataset was assembled from the clinical and molecular data of the TCGA-LUAD PanCancer Atlas project (*The Cancer Genome Atlas – Lung Adenocarcinoma*), combining each patient's overall survival time, vital status, age at diagnosis, sex, pathological stage, histological subtype and molecular indicators associated with tumor hypoxia.

The specific objective of the analysis is to evaluate whether, and to what extent, patients' clinical and molecular characteristics are associated with overall survival. In particular, the study aims to assess whether biological markers related to tumor hypoxia provide additional prognostic information beyond the main clinical features, allowing the identification of profiles associated with different survival outcomes. Survival time is therefore analyzed using Kaplan–Meier curves, the log-rank test and the semiparametric Cox model. The main model assumptions are also assessed using appropriate diagnostic tools.

2 Dataset

2.1 Dataset construction

The data come from the TCGA-LUAD PanCancer Atlas study and were obtained through cBioPortal. The original clinical dataset contains 566 patients with lung adenocarcinoma and 40 variables. The dataset has a patient-level structure: each row represents one patient and no duplicate identifiers are present.

Survival observations in this study are *right-censored*. For patients who died, the elapsed time to death is known, whereas for patients still alive, observation ends at the last available contact. Kaplan–Meier curves, log-rank tests and Cox

models assume censoring to be independent and non-informative with respect to the risk of death.

After checking the outcome, removing zero survival times and excluding observations with missing values in the covariates used, the common analysis sample comprises 485 patients, of whom 177 died and 308 were censored. The following information was considered for each patient:

- **Overall survival time:** number of months from diagnosis to death or, if the event was not observed, to the last available contact. Time is accompanied by a binary indicator taking the value 1 for death and 0 for censoring.
- **Age at diagnosis:** the patient's age at diagnosis, in years. In the Cox models, age was rescaled in ten-year units to make the hazard ratio easier to interpret.
- **Sex:** categorical variable with Female and Male categories.
- **Pathological stage:** classification of disease extent according to the American Joint Committee on Cancer (AJCC) system. Original categories IA and IB were combined into stage I, IIA and IIB into stage II, and IIIA and IIIB into stage III; stage IV was retained separately.
- **Histological subtype:** variable derived from the ICD-O-3 histology code and grouped into Adenocarcinoma NOS, Mixed adenocarcinoma and Other subtype.
- **Molecular indicators of tumor hypoxia:** composite scores derived from the expression of predefined gene sets involved in the cellular response to reduced oxygen availability. They do not directly measure the oxygen concentration within the tumor, but summarize the extent to which the sample's gene-expression profile is consistent with a hypoxic phenotype. Higher values indicate, in relative terms, greater activation of the molecular hypoxia signature. Buffa Hypoxia Score, Ragnum Hypoxia Score and Winter Hypoxia Score, calculated from three different gene signatures, were available in the dataset.

The main variables are summarized in Table 1.

Table 1: Main variables considered in the analysis.

Variable	**Type**	**Description**
Survival time	Continuous	Overall survival or follow-up time, in months.
Event	Binary	Event indicator: 1 for death and 0 for censoring.
Age	Continuous	Age at diagnosis, in years.
Sex	Categorical	Patient sex, categorized as Female or Male.
Pathological stage	Categorical	Disease extent grouped into stages I, II, III and IV.
Histological subtype	Categorical	Subtype grouped into Adenocarcinoma NOS, Mixed adenocarcinoma and Other subtype.
Hypoxia score	Continuous	Composite gene-expression indices associated with a hypoxic tumor phenotype.

Table 2 summarizes the progression from the original dataset to the sample used in the multivariable models.

Table 2: Construction of the analytical sample.

Step	**Number of patients**
Original clinical dataset	566
Complete survival outcome	505
Valid outcome and survival time greater than zero	501
Common dataset for Cox models	485

2.2 Missing data

Missing values were examined separately for the survival outcome and for the covariates considered in the analysis. Of the 566 patients in the original dataset, 61 lacked complete information on survival time or status and could not be included. Among the 505 patients with a complete outcome, four had an observed time of zero and were excluded, leaving 501 patients with a valid outcome.

Table 3 reports missing values in the main covariates among these patients.

Table 3: Missing values in the main covariates of the survival dataset.

Variable	**Missing values**	**Percentage**
Age at diagnosis	10	2,0%
Sex	0	0,0%
Pathological stage	2	0,4%
Histological subtype	0	0,0%
Buffa Hypoxia Score	4	0,8%
Ragnum Hypoxia Score	4	0,8%
Winter Hypoxia Score	4	0,8%

Because the overall proportion of missing covariate data was small, no imputation procedure was performed. A complete-case analysis was adopted to ensure that all multivariable models were estimated on the same group of patients and that comparisons between models could be interpreted directly.

Only patients with available survival time, event status, age, sex, pathological stage, histological subtype and the hypoxia score selected for the model were therefore retained. The common Cox analysis dataset contains 485 patients, corresponding to 96.8% of the 501 patients with a valid outcome.

The information lost by excluding incomplete cases is therefore limited. Nevertheless, complete-case analysis assumes that missing data do not introduce important systematic differences between included and excluded patients; this is a limitation when interpreting the results.

3 Exploratory analysis

3.1 Variables

The exploratory analysis was performed on the common Cox-model dataset of 485 patients. The distributions of observed survival time, age at diagnosis, the main categorical variables and the molecular tumor-hypoxia indicators were examined.

3.1.1 Survival time

Figure 1 shows the distribution of overall survival or follow-up time, in months.

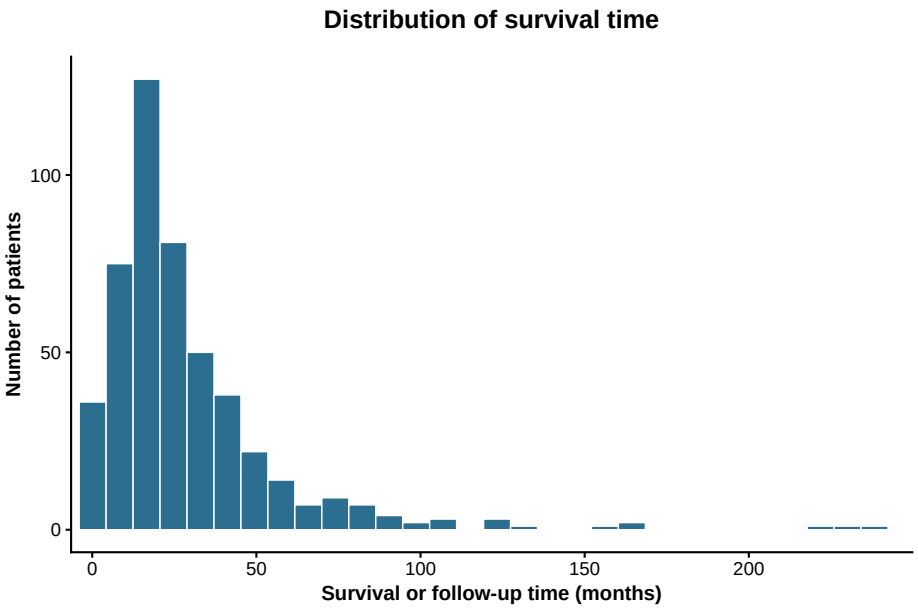


Figure 1: Distribution of survival or follow-up time.

Observed time has a median of 21.4 months and an interquartile range of 13.7 to 36.1 months; the mean is 29.1 months. The distribution is strongly right-skewed: most patients have relatively short observed times, while a small number are followed for much longer periods, up to 238.3 months.

3.1.2 Age at diagnosis

The distribution of patients' age at diagnosis is shown in Figure 2.

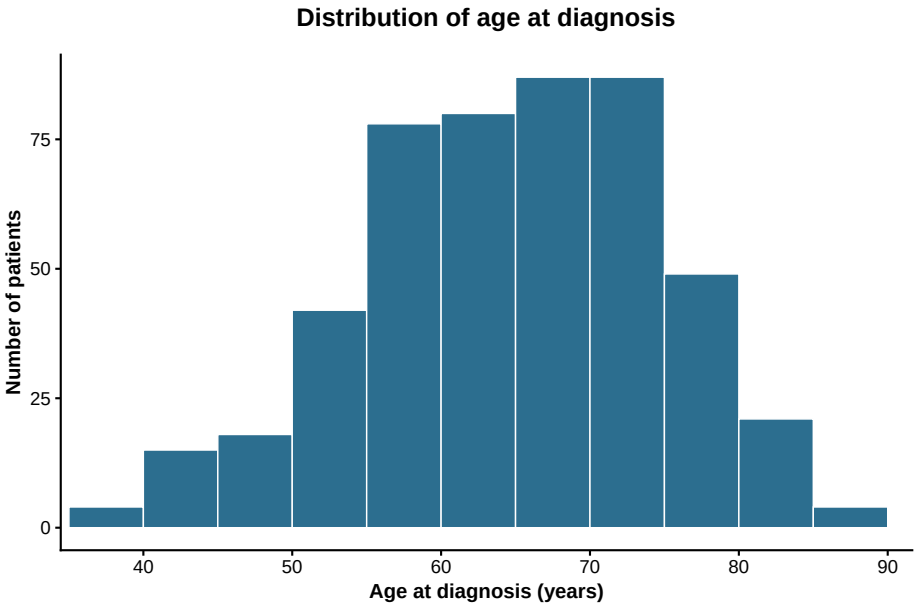


Figure 2: Distribution of age at diagnosis.

Age is mainly concentrated between approximately 55 and 80 years, with few very young or particularly old patients. The distribution shows no anomalies that would prevent age from being used as a quantitative covariate in the Cox model.

3.1.3 Categorical variables

Sex, pathological stage and histological subtype distributions are shown separately in Figures 3, 4 and 5.

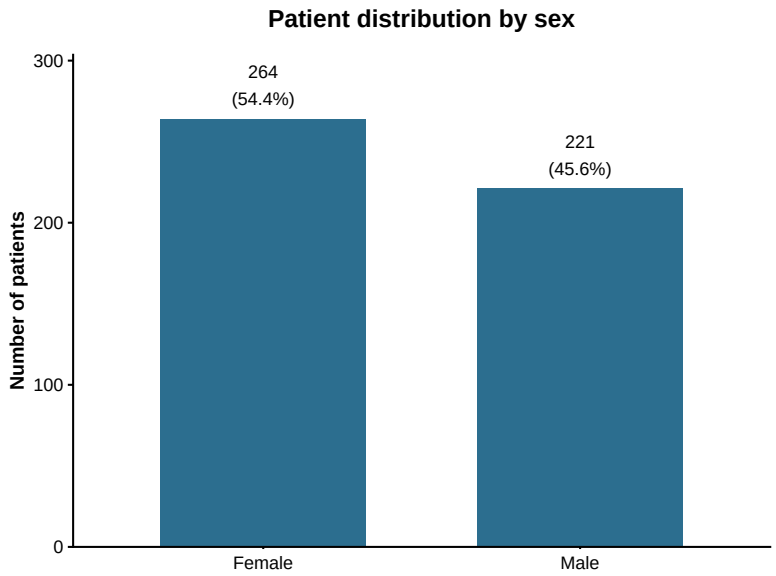


Figure 3: Distribution of patients by sex.

There are 264 women (54.4% of the sample) and 221 men (45.6%).

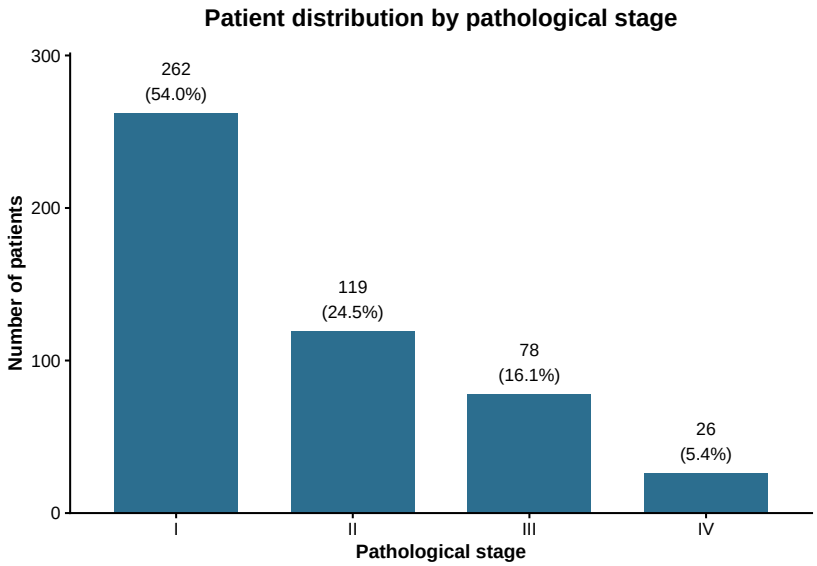


Figure 4: Distribution of patients by pathological stage.

The distribution is skewed toward earlier stages: more than half of patients have stage I disease, while stage IV accounts for only 5.4% of the sample. Estimates for stage IV therefore require greater caution.

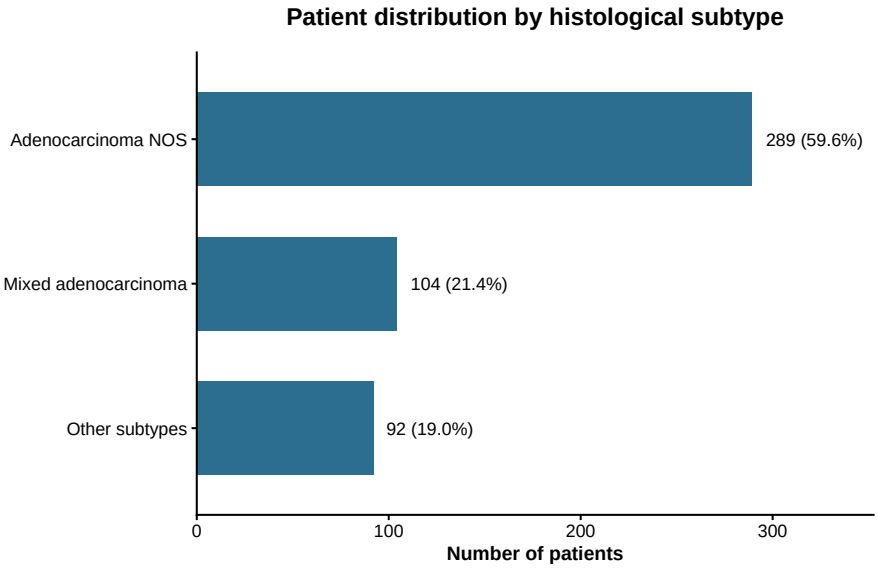


Figure 5: Distribution of patients by histological subtype.

Adenocarcinoma NOS is the predominant category, accounting for approximately 60% of the sample. The Other subtypes category is smaller and combines different histological forms, making it internally heterogeneous.

3.1.4 Tumor-hypoxia indicators

The dataset contains three tumor-hypoxia scores: Buffa Hypoxia Score, Ragnum Hypoxia Score and Winter Hypoxia Score. Each summarizes expression of a different gene signature responding to oxygen deprivation. These values are therefore not direct measurements of tissue oxygenation but relative molecular indices: a higher score indicates a tumor expression profile more consistent with a hypoxic response. Because the signatures use different gene sets and measurement scales, their absolute values should not be compared directly. Their respective distributions are shown in Figure 6.

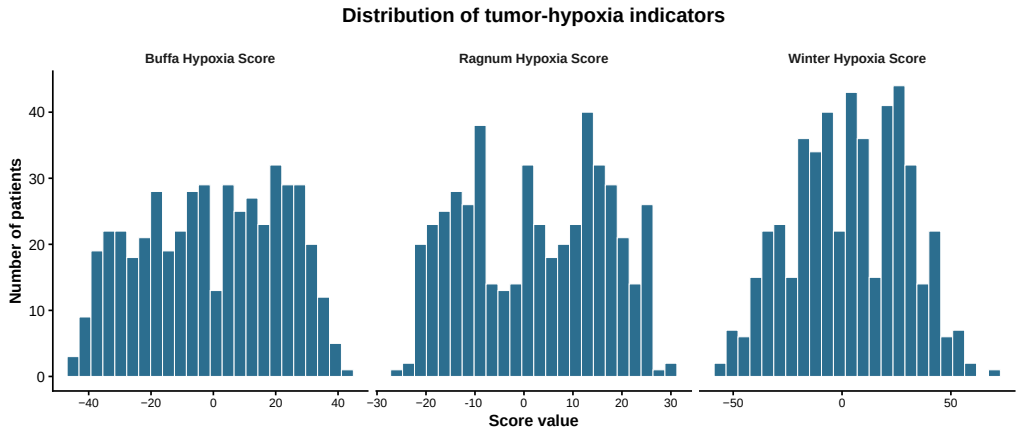


Figure 6: Distribution of molecular tumor-hypoxia indicators.

All three scores take both negative and positive values and have fairly regular distributions without extreme tails requiring preliminary transformations. They are therefore initially treated as continuous quantitative variables in the models. Although measured on different scales, their linear associations can be compared using Pearson's correlation coefficient.

3.2 Correlations

The linear relationships among the three hypoxia indicators were assessed using Pearson's correlation coefficient. The matrix is shown in Figure 7.

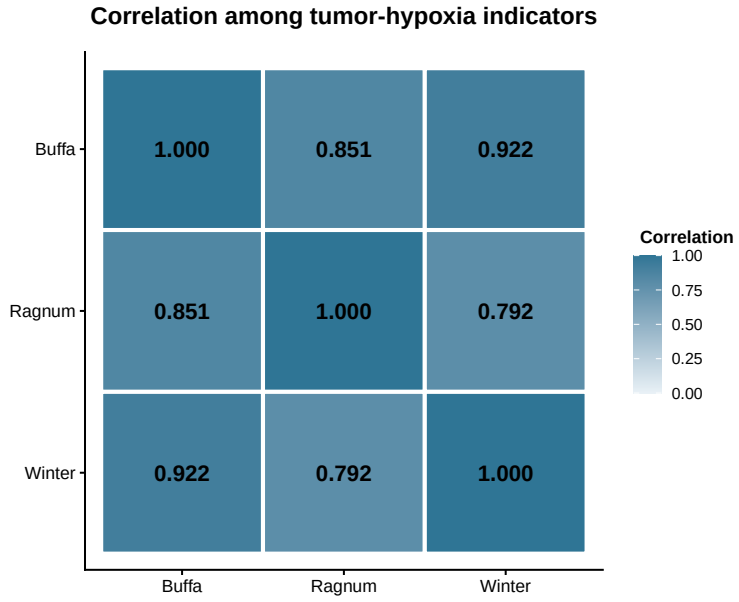


Figure 7: Correlation matrix of tumor-hypoxia indicators.

Correlations are high: the Buffa–Ragnum correlation is 0.851, Buffa–Winter is 0.922 and Ragnum–Winter is 0.792.

Table 4: Correlations among tumor-hypoxia indicators.

Indicator 1	Indicator 2	Correlation
Buffa	Ragnum	0,851
Buffa	Winter	0,922
Ragnum	Winter	0,792

The strong association among the three scores suggests that they describe substantially overlapping components of the same biological phenomenon. Including them simultaneously in the model could therefore introduce collinearity and make estimates unstable. To keep the model parsimonious and interpretable, one indicator was selected: Buffa Hypoxia Score was used as the primary signature. Ragnum and Winter were used to examine the consistency of the molecular phenomenon, but were not entered into the model together. This is an operational choice, not evidence that the Buffa score is formally superior to the other two indicators.

4 Survival analysis

The survival function was estimated using the Kaplan–Meier method. Time was defined as the interval, in months, between diagnosis and death or, for

patients without an observed event, between diagnosis and the last available contact. Patients alive at the end of follow-up were therefore treated as censored observations.

4.1 Overall Kaplan–Meier curve

The overall curve describes how survival probability changes in the study population. As shown in Figure 8, estimated survival probability decreases progressively during follow-up, with a more pronounced decline in the early years and a subsequent tail with fewer patients remaining at risk.

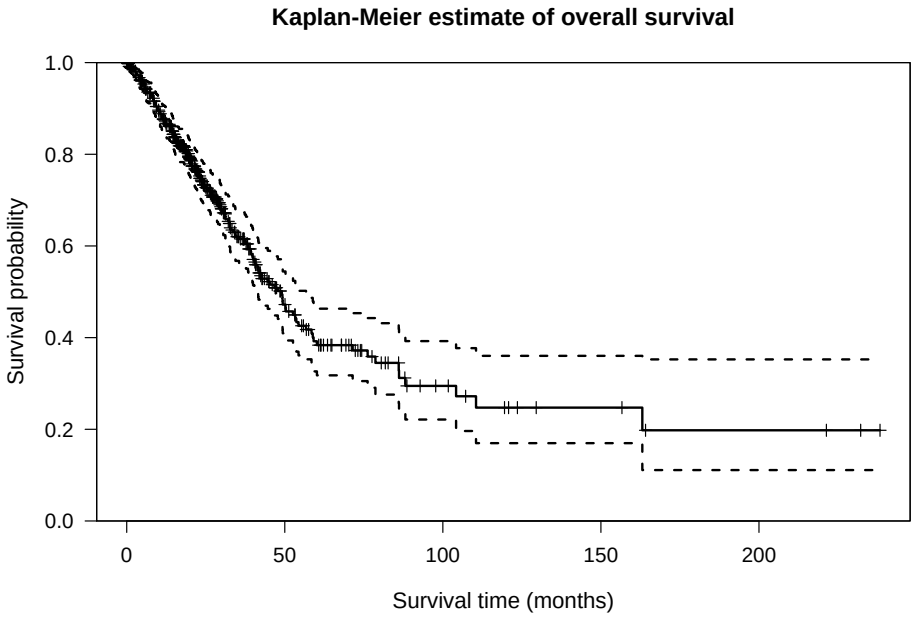


Figure 8: Overall Kaplan–Meier curve with a 95% confidence interval.

Estimated median survival is 49.1 months, with a 95% confidence interval of 41.4 to 56.7 months. Estimated survival probabilities are 87.5% at 12 months, 61.5% at 36 months and 39.2% at 60 months.

4.2 Kaplan–Meier curves by pathological stage

To assess the association between disease extent and survival, separate curves were estimated for the four pathological stages, as shown in Figure 9.

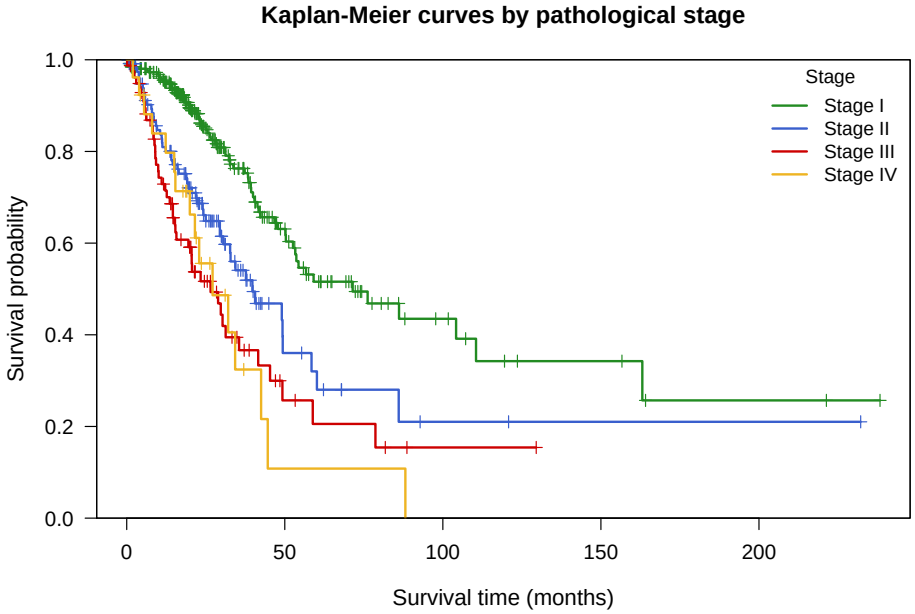


Figure 9: Kaplan–Meier curves by pathological stage.

The stage I curve generally remains above the other curves, indicating a more favorable prognosis. Stages II, III and IV show progressively lower survival probabilities, with overall worse outcomes in the more advanced stages.

The log-rank test indicates a statistically significant difference among the four groups' survival curves:

$$p < 0{,}001.$$

Pathological stage is therefore strongly associated with overall survival in the unadjusted analysis. However, this comparison does not account for the patients' other clinical and molecular characteristics, which will subsequently be considered in the Cox model.

4.3 Multivariable and stratified Cox models

To assess simultaneously the associations between patients' clinical and molecular characteristics and the risk of death, a multivariable proportional-hazards Cox model was first estimated. It includes age at diagnosis, sex, pathological stage, histological subtype and Buffa Hypoxia Score.

Age was expressed in ten-year units and Buffa Hypoxia Score in ten-point units to make the corresponding hazard ratios easier to interpret. Reference categories were female sex, stage I and adenocarcinoma NOS.

The initial model can be written as:

$$h(t \mid X) = h_0(t) \exp\{\beta_1 \text{Age}_{10} + \beta_2 \text{Sex} + \beta_3 \text{Stage} + \beta_4 \text{Histology} + \beta_5 \text{Buffa}_{10}\}. \tag{1}$$

Results from the standard multivariable Cox model are reported in Table 5.

Table 5: Standard multivariable Cox model.

Covariate	**HR**	**95% CI**	***p*-value**
Age, per 10 years	1,236	1,052–1,452	0,010
Male vs Female	1,013	0,750–1,367	0,933
Stage II vs I	1,931	1,334–2,795	$< 0,001$
Stage III vs I	2,888	1,960–4,255	$< 0,001$
Stage IV vs I	3,821	2,170–6,729	$< 0,001$
Mixed adenocarcinoma vs NOS	0,748	0,504–1,111	0,150
Other subtypes vs NOS	0,477	0,282–0,807	0,006
Buffa score, per 10 points	1,117	1,036–1,206	0,004

The model is globally significant according to the likelihood ratio test ($p < 0,001$).

Holding the other covariates constant, a ten-year increase in age is associated with approximately a 23.6% increase in the hazard of death. Sex, however, does not show a statistically significant association with the event.

Pathological stage is strongly associated with prognosis. Compared with stage I, the hazard is approximately 1.9 times greater in stage II, 2.9 times greater in stage III and 3.8 times greater in stage IV.

Regarding histology, mixed adenocarcinoma does not differ significantly from the NOS reference category. Other subtypes, in contrast, have a lower hazard, with an HR of 0.477. Finally, a ten-point increase in Buffa Hypoxia Score is associated with an 11.7% increase in the hazard of death.

4.3.1 Checking the proportional-hazards assumption

The proportional-hazards assumption was assessed with the test based on Schoenfeld residuals. Results for the standard model are reported in Table 6.

Table 6: Schoenfeld test for the standard multivariable Cox model.

Covariate	**Chi-square**	**df**	***p*-value**
Age	1,611	1	0,204
Sex	5,021	1	0,025
Pathological stage	10,313	3	0,016
Histological subtype	0,445	2	0,801
Buffa Hypoxia Score	1,461	1	0,227
Global	**18,719**	**8**	**0,016**

The test shows no violation for age, histological subtype or Buffa Hypoxia Score. In contrast, sex and pathological stage have p-values below 0.05. The global test is also significant, indicating that the standard model does not fully satisfy the proportional-hazards assumption. For sex, a non-significant average hazard ratio does not rule out variation of its effect over time.

Figure 10 displays only the Schoenfeld residuals for sex and pathological stage, the two covariates for which the test suggests possible time variation in effect.

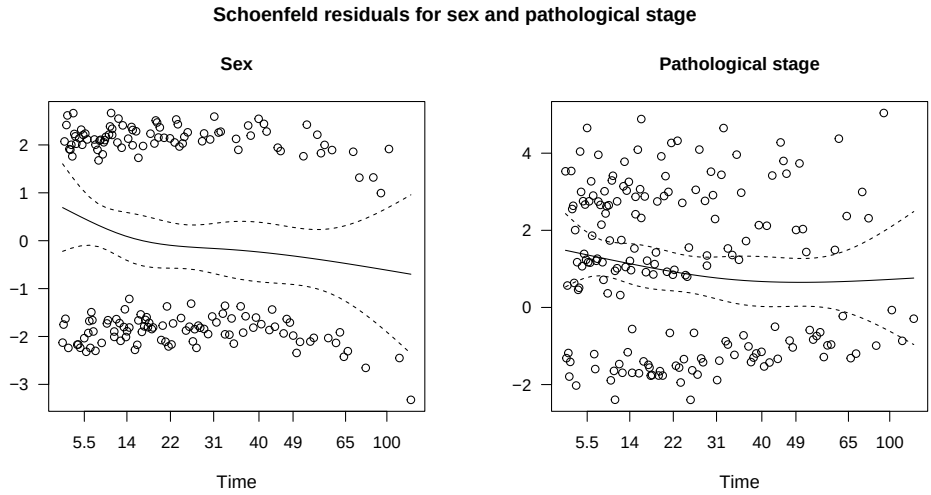


Figure 10: Schoenfeld residuals for sex and pathological stage.

The plots support the finding that the effects of sex and pathological stage are not constant throughout follow-up. In particular, the smoothed curves of the estimated coefficients are not perfectly horizontal over time, consistent with the significant Schoenfeld tests.

Because the proportional-hazards assumption is not met for these two covariates, a Cox model stratified by sex and pathological stage was fitted. This allows each combination of the two variables to have its own baseline hazard function, without imposing a time-constant effect.

4.3.2 Stratified Cox model

To address the proportionality violation, a Cox model jointly stratified by pathological stage and sex was estimated. Stratification allows each stage–sex combination to have its own baseline hazard function, without requiring the effects of these two variables to remain constant over follow-up.

The final model is:

$$h_s(t \mid X) = h_{0s}(t) \exp\{\beta_1 \text{Age}_{10} + \beta_2 \text{Histology} + \beta_3 \text{Buffa}_{10}\}, \tag{2}$$

where s identifies the stratum defined by the combination of pathological stage and sex. Stratification therefore permits different baseline hazard functions across the observed combinations, up to a theoretical maximum of eight strata. The trade-off is that the model does not return a single hazard ratio for sex or pathological stage, because their effects are absorbed into the stratum-specific baseline hazard functions.

Results from the final stratified model are reported in Table 7.

Table 7: Cox model stratified by pathological stage and sex.

Covariate	**HR**	**95% CI**	***p*-value**
Age, per 10 years	1,241	1,055–1,460	0,009
Mixed adenocarcinoma vs NOS	0,769	0,516–1,144	0,195
Other subtypes vs NOS	0,473	0,275–0,812	0,007
Buffa score, per 10 points	1,107	1,025–1,196	0,010

The stratified model is globally significant according to the likelihood ratio test ($p < 0,001$).

Even after allowing stage and sex to have different baseline hazards, age and Buffa Hypoxia Score remain significantly associated with survival. A ten-year increase in age corresponds to an approximately 24.1% increase in hazard, while a ten-point increase in the Buffa score is associated with a 10.7% increase in risk.

Other histological subtypes continue to have a lower hazard than adenocarcinoma NOS, whereas mixed adenocarcinoma does not show a statistically significant difference.

4.3.3 Assessment of the stratified model

The Schoenfeld test was repeated for the covariates estimated in the stratified model. Results are presented in Table 8.

Table 8: Schoenfeld test for the stratified Cox model.

Covariate	**Chi-square**	**df**	***p*-value**
Age	2,796	1	0,095
Histological subtype	0,167	2	0,920
Buffa Hypoxia Score	0,191	1	0,662
Global	**3,613**	**4**	**0,461**

All p-values exceed 0.05 and the global test is not significant. There is thus no evidence of a remaining proportional-hazards violation. For this reason, no additional Schoenfeld-residual plots for the stratified model are presented.

The stratified model is therefore regarded as the final model, as it accommodates the non-proportionality of sex and stage while retaining the associations with age, histological subtype and tumor hypoxia. This choice nevertheless means that no single hazard ratio is estimated for the two stratification variables.

4.4 Diagnostics of the final model

To assess the adequacy of the functional form of the continuous covariates in the final stratified model, Martingale residuals were examined against age and Buffa Hypoxia Score.

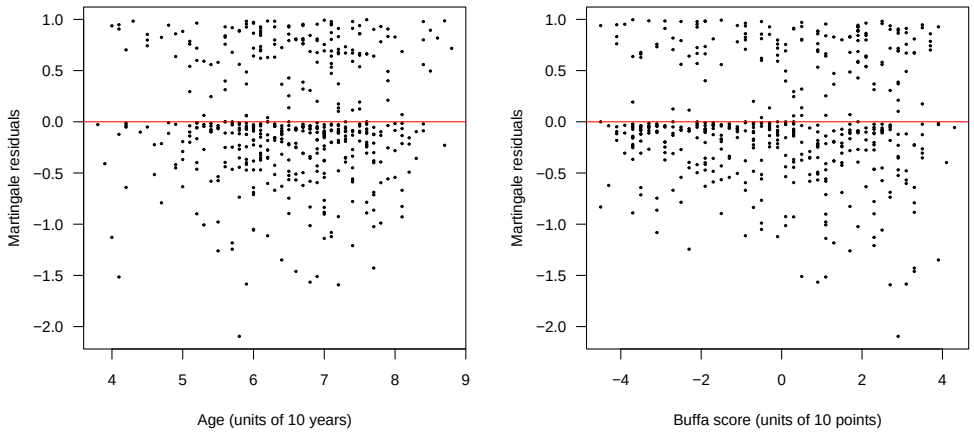


Figure 11: Martingale residuals versus age and Buffa Hypoxia Score.

The horizontal red line marks the reference value of zero. In both plots, residuals are scattered above and below that line, without particularly pronounced systematic patterns against age or Buffa Hypoxia Score. The graphical inspection therefore does not suggest obvious departures from the linear functional form used in the model.

4.5 Clinical–molecular prognostic profiles

To illustrate the joint effects of age and tumor hypoxia, three hypothetical profiles were defined. The favorable profile represents a 55-year-old female patient with a Buffa Hypoxia Score of −20, the intermediate profile a 65-year-old female patient with a score of 0, and the unfavorable profile a 75-year-old female patient with a score of 20. Female sex, stage I and adenocarcinoma NOS were held constant across all three profiles.

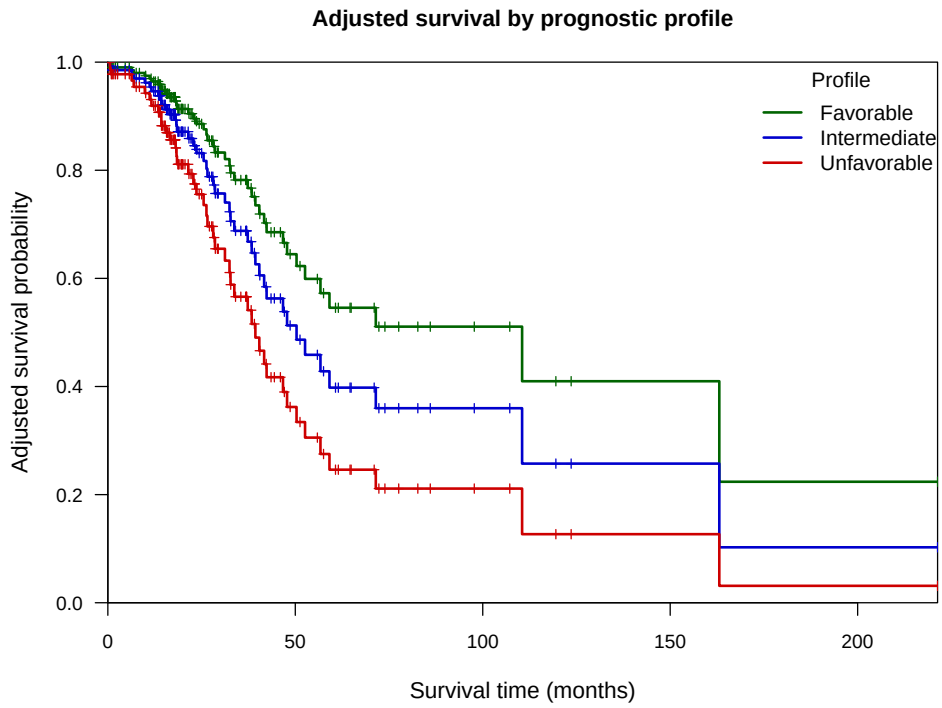


Figure 12: Estimated survival curves for the three clinical–molecular prognostic profiles.

The curves show a clear prognostic gradient. At 60 months, estimated survival is 0.546 for the favorable profile, 0.398 for the intermediate profile and 0.246 for the unfavorable profile. The separation reflects the combined effects of age and Buffa Hypoxia Score under the fitted model.

5 Study limitations

The results should be interpreted in light of several limitations. First, the analysis uses observational, retrospective data, so the estimated associations cannot be interpreted causally. A complete-case analysis was also adopted: the proportion of excluded observations is small, but valid estimation requires that missing values do not introduce important systematic differences between included and excluded patients.

The number of stage IV patients is small, and jointly stratifying by stage and sex further divides the sample into smaller groups. Estimates for sparsely represented strata should therefore be interpreted cautiously. Finally, the Other subtypes category combines different histological forms and may conceal considerable internal heterogeneity.

6 Conclusions

The analysis shows that age at diagnosis, Buffa Hypoxia Score and histological subtype are associated with survival in patients with lung adenocarcinoma. In particular, a ten-year increase in age corresponds to an approximately 24% increase in hazard, while a ten-point increase in the Buffa score is associated with an increase of about 11%.

Pathological stage is strongly associated with prognosis in the standard model. Because sex and stage do not satisfy the proportional-hazards assumption, a stratified model was used; this removes the evidence of remaining violations while retaining the associations with the other covariates.

Overall, the findings indicate that combining clinical characteristics with tumor-hypoxia information can distinguish different prognostic levels, subject to limitations arising from the observational nature of the data and small sizes of some subgroups.

7 Sources

The data used in this analysis come from the study *Lung Adenocarcinoma (TCGA, PanCancer Atlas)*, disponibile su cBioPortal for Cancer Genomics:

`https://www.cbioportal.org/study/summary?id=luad_tcga_pan_can_atlas_2018`